Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A ray of light, travelling in a medium of refractive index μ, is incident at an angle i on a composite transparent plate consisting of three plates of refractive indices μ_1, μ_2 and μ_3. The ray emerges from the composite plate into a medium of refractive index μ', at angle x. Then

(A) $\sin x = \sin i$

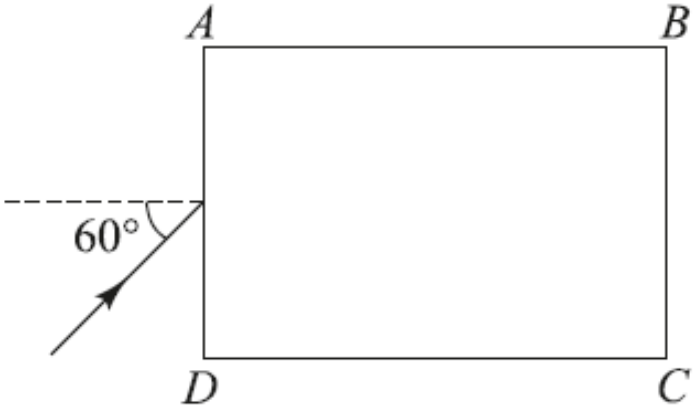
(B) $\sin x = \frac{\mu}{\mu'} \sin i$

(C) $\sin x = \frac{\mu'}{\mu} \sin i$

(D) $\sin x = \frac{\mu_1}{\mu_2} \cdot \frac{\mu_3}{\mu_2} \cdot \frac{\mu}{\mu'} \sin i$

Q.2

A ray of light is incident at an angle of 60° on face AD of a transparent slab. What should be the refractive index of the material of the slab so that total internal reflection occurs at face AB of the slab? [see Fig.]



(A) 1.5

(B) $\sqrt{3}$

(C) $\frac{2\sqrt{5}}{3}$

(D) $\frac{\sqrt{7}}{2}$

Q.3

A scuba diver in an empty swimming pool uses a magnifier $(n = 1.25)$ to enlarge the print on a plastic instruction sheet. If the pool is filled with water $(n = 1.33)$, what happens to the magnification of the print?

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(A) It increases and is greater than one.

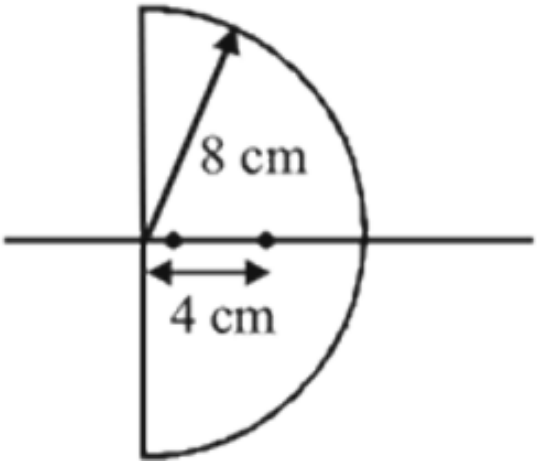
(B) It stays the same.

(C) It decreases, but is still greater than one.

(D) It decreases and is less than one

Q.4

A plastic hemisphere has a radius of curvature of $8\ \mathrm{cm}$ and an index of refraction of 1.6. On the axis halfway between the plane surface and the spherical one $(4\ \mathrm{cm}$ from each) is a small object O. The distance between the two images when viewed along the axis from the two sides of the hemisphere is approximately



(A) $1.0\ \mathrm{cm}$

(B) $1.5\ \mathrm{cm}$

(C) $3.75\ \mathrm{cm}$

(D) $2.5\ \mathrm{cm}$

Q.5

For an optical fibre consists of a core (refractive index $= \mu_1$) and cladding (refractive index $= \mu_2$):

(A) $\mu_1 > \mu_2$

(B) $\mu_1 < \mu_2$

(C) $\mu_1 = \mu_2$

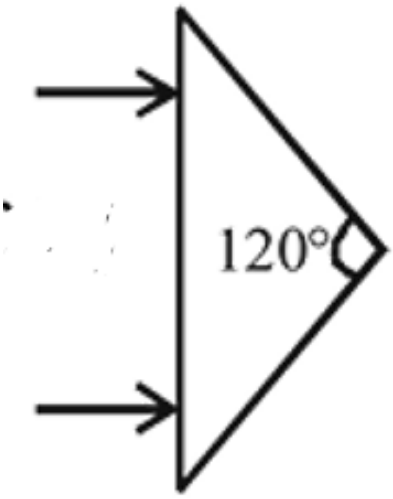
(D) $\frac{\mu_1}{\mu_2} = \frac{1}{2}$

Q.6

An isosceles prism of angle 120° has a refractive index 1.44. Two parallel monochromatic rays enter the

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prism parallel to each other in air as shown. The rays emerge from the opposite faces



(A) are parallel to each other

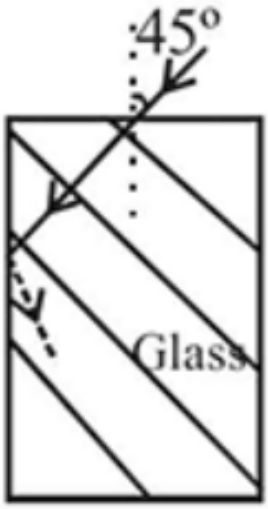
(B) are diverging

(C) make an angle $2\left[\sin^{-1}(0.72) - 30^{\circ}\right]$ with each other

(D) make an angle $2\sin^{-1}(0.72)$ with each other

Q.7

A light ray falls on a rectangular glass slab as shown. The index of refraction of the glass, if total internal reflection is to occur at the vertical face, is



(A) $\sqrt{3/2}$

(B) $\frac{(\sqrt{3}+1)}{2}$

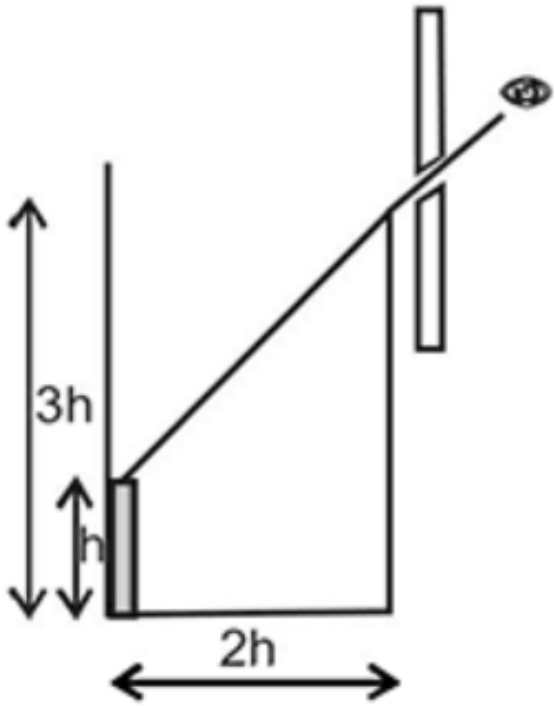
(C) $\frac{(\sqrt{2}+1)}{2}$

(D) $\sqrt{5}/2$

Q.8

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An observer can see through a pinhole the top end of a thin rod of height h, placed as shown in the figure. The beaker height is $3\ \mathrm{h}$ and its radius h. When the beaker is filled with a liquid up to a height $2\ \mathrm{h}$, he can see the lower end of the rod. Then the refractive index of the liquid is



(A) $\frac{5}{2}$

(B) $\sqrt{\frac{5}{2}}$

(C) $\sqrt{\frac{3}{2}}$

(D) $\frac{3}{2}$

Q.9

A rectangular block of glass is placed on a mark made on the surface of the table and it is viewed from the vertical position of eye. If refractive index of glass be μ and its thickness d, then the mark will appear to be raised up by

(A) $\frac{(\mu+1)\mathrm{d}}{\mu}$

(B) $\frac{(\mu-1)\mathrm{d}}{\mu}$

(C) $\frac{(\mu+1)}{\mu\mathrm{d}}$

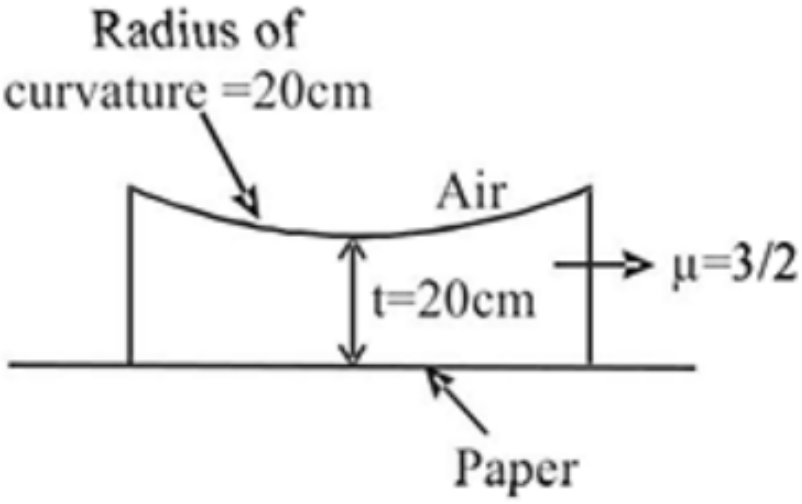
(D) $\frac{(\mu-1)\mu}{\mathrm{d}}$

Q.10

A planoconcave lens is placed on a paper on which a flower is drawn. How far above its actual position

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does the flower appear to be?



(A) 10 cm

(B) 15 cm

(C) 50 cm

(D) None of these

Q.11

A fish looking up through the water sees the outside world contained in a circular horizon. If the refractive index of
water is $\frac{4}{3}$ and the fish is 12 cm below the surface, then
what is the radius of this circle in cm ?

(A) 13.5

(B) 13.8

(C) 13.4

(D) 13.6

Q.12

When a plane mirror is placed horizontally on a level ground at a distance of 60 m from the foot of a tower, the top of the tower and its image in the mirror subtend an angle of 90° at the eye. What is the height of the tower in metre?

(A) 50

(B) 40

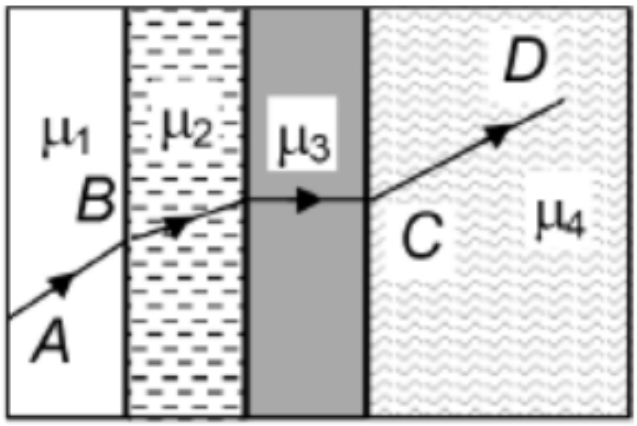
(C) 65

(D) 60

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Q.13

A ray of light passes through four transparent media with refractive indices μ_1, μ_2, μ_3 and μ_4 as shown in the figure. The surface of all media are parallel. If the emergent ray CD is parallel to the incident ray AB, then



(A) $\mu_1 = \mu_2$

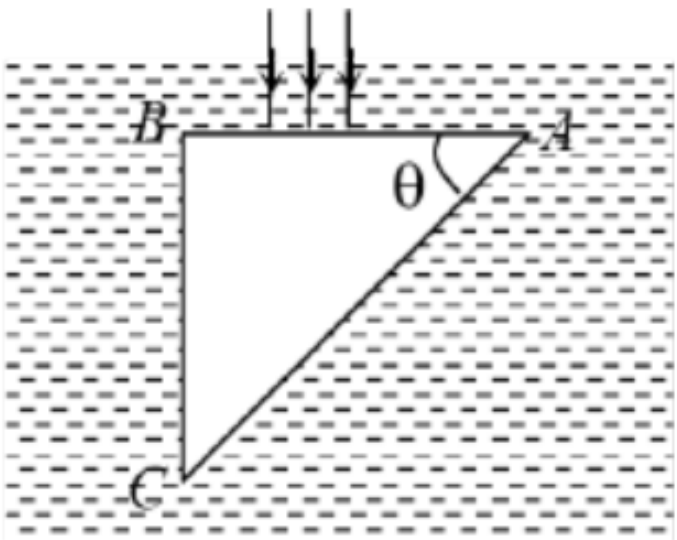
(B) $\mu_2 = \mu_3$

(C) $\mu_3 = \mu_4$

(D) $\mu_4 = \mu_1$

Q.14

A glass prism of refractive index 1.5 is immersed in water $(\mu = 4/3)$. A light beam incident nomally on the face AB is totally reflected to reach the face BC, if



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(A) $\sin\theta \geq \frac{8}{9}$

(B) $\sin\theta \leq \frac{2}{3}$

(C) $\sin\theta = \frac{4}{5}$

(D) $\frac{2}{3} < \sin\theta < \frac{8}{9}$

Q.15

A bulb is placed at a depth of $2\sqrt{7}$ m in water and a floating opaque disc is placed over the bulb so that the bulb is not visible from the surface. The radius of the disc should be
at least $(\mu_{\text{water}} = 4/3)$

(A) 42 m

(B) 6 m

(C) $2\sqrt{7}$ m

(D) 12 m

Answers & Solutions

Q.1 Answer:

$\sin x = \frac{\mu}{\mu'}\sin i$

Solution:

Refer to Fig.

Applying Snell's law to successive refractions, we have

$\frac{\sin i}{\sin r_1} = \frac{\mu_1}{\mu}$; $\frac{\sin r_1}{\sin r_2} = \frac{\mu_2}{\mu_1}$

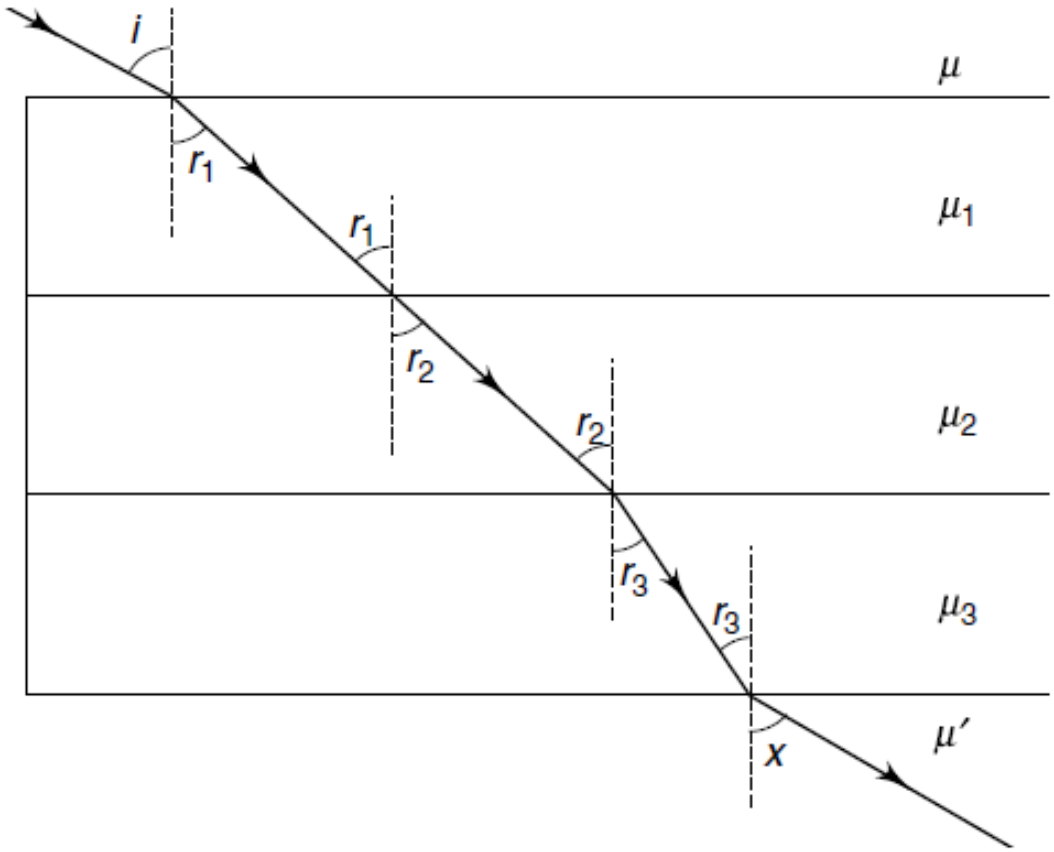
$\frac{\sin r_2}{\sin r_3} = \frac{\mu_3}{\mu_2}$; $\frac{\sin r_3}{\sin x} = \frac{\mu'}{\mu_3}$

Multiplying these equations we get

$\frac{\sin i}{\sin x} = \frac{\mu'}{\mu}$

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or $\sin x = \frac{\mu}{\mu'} \sin i$



Q.2 Answer:

$\frac{\sqrt{7}}{2}$

Solution:

Refer to Fig. From Snell's law,

$\mu \sin r = \sin 60^\circ = \frac{\sqrt{3}}{2}$

$\sin r = \frac{\sqrt{3}}{2\mu}$ **(1)**

The refracted ray will be totally reflected at E **if it is incident at an angle equal to or greater than the critical angle** i_c **which is given by**

$\sin i_c = \frac{1}{\mu}$

It follows from the figure that $r = 90^\circ - i_c$**. Therefore,**

$$\sin r = \sin (90^\circ - i_c) = \cos i_c$$

$$\Rightarrow \quad \sin r = \left(1 - \sin^2 i_c\right)^{1/2}$$

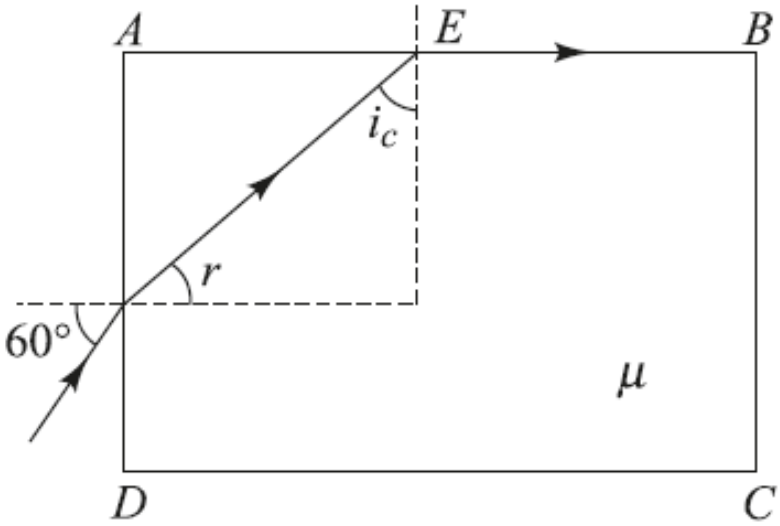
Using (1) and (2) in (3), we get

$\frac{\sqrt{3}}{2\mu} = \left(1 - \frac{1}{\mu^2}\right)^{1/2}$

$\Rightarrow \quad \frac{3}{4\mu^2} = 1 - \frac{1}{\mu^2}$

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$\Rightarrow \quad \mu = \frac{\sqrt{7}}{2}, \quad$ **which is choice** (d)



Q.3 Answer:

It decreases and is less than one

Solution:

Since the water has a greater index of refraction than the lens, the magnifier now acts like a diverging lens, resulting in an image that is smaller than the object.

Q.4 Answer:

2.5 cm

Solution:

Distance of image from the plane surface is as follows:

$x_1 = \frac{4}{1.6} = 2.5 \text{ cm} \left(d_{\text{app}} = \frac{d_{\text{actual}}}{\mu} \right)$

For the curved side

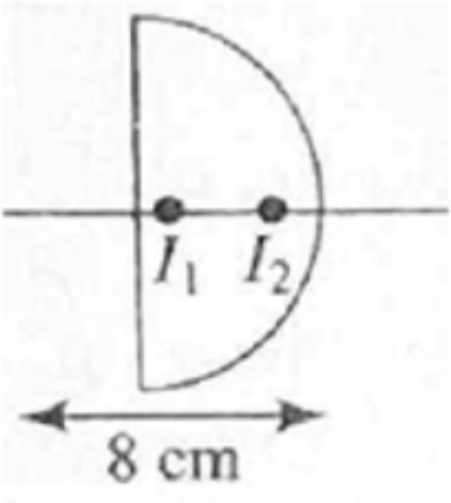
$$\frac{1.6}{4} + \frac{1}{x_2} = \frac{1 - 1.6}{8}$$

$\therefore \quad x_2 \approx -3.0 \text{ cm}$

The minus sign means the image is on the object side.

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$\therefore \quad \mathrm{I_1I_2} = (8 - 2.5 - 3.0)\mathrm{cm} = 2.5\ \mathrm{cm}$



Q.5 Answer:

$\mu_1 > \mu_2$

Solution:

No solution available

Q.6 Answer:

make an angle $2\left[\sin^{-1}(0.72) - 30^\circ\right]$ with each other

Solution:

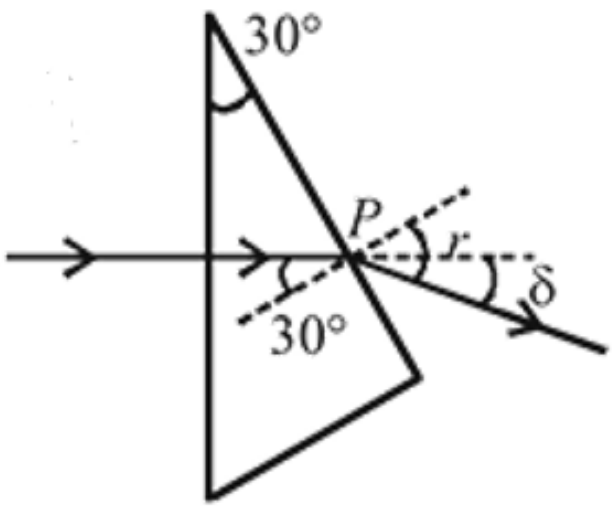
Applying Snell's law at P ,

$\mu = \frac{\sin r}{\sin 30^\circ}$

$\sin r = \frac{1.44}{2} = 0.72$

$\therefore \quad \delta = r - 30^\circ = \sin^{-1}(0.72) - 30^\circ$

$\therefore$ **The rays make an angle of** $2\delta = 2\left[\sin^{-1}(0.72) - 30^\circ\right]$ **with each other .**



Q.7 Answer:

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$\sqrt{3/2}$

Solution:

For point A, $_a\mu_g = \frac{\sin 45^\circ}{\sin r}$

$\Rightarrow \sin r = \frac{1}{\sqrt{2}_a\mu_g}$

For point B, $\sin(90^\circ - r) = {}_g\mu_a$ **where,**

$(90^\circ - r)$ **is critical angle.**

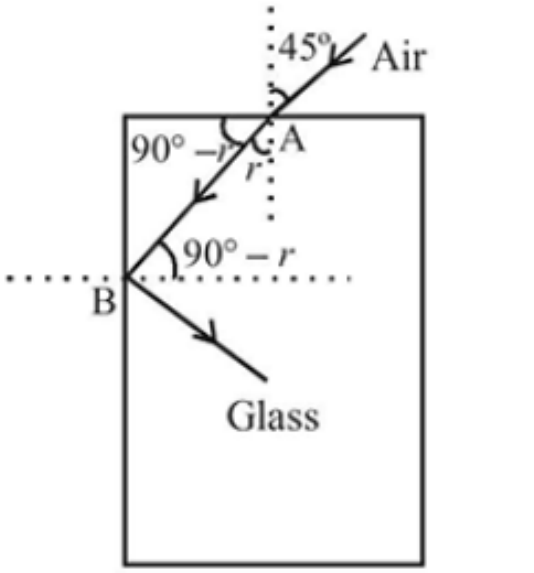
$\therefore \cos r = {}_g\mu_a = \frac{1}{{}_a\mu_g}$

$\Rightarrow {}_a\mu_g = \frac{1}{\cos r}$

$= \frac{1}{\sqrt{1-\sin^2 r}} = \frac{1}{\sqrt{1-\frac{1}{2_a\mu_g^2}}}$

$\Rightarrow {}_a\mu_g{}^2 = \frac{1}{1-\frac{1}{2_a\mu_g^2}} = \frac{2_a\mu_g^2}{2_a\mu_g^2-1}$

$\Rightarrow 2_a\mu_g^2 - 1 = 2 \Rightarrow a\mu_g = \sqrt{\frac{3}{2}}$



Q.8 Answer:

$\sqrt{\frac{5}{2}}$

Solution:

For the image of poin P **to be seen by the observer, it should be formed at point** Q**. In** ΔQNS,

$NS = QS = 2h$

$\therefore \angle NQS = 45^\circ$

$\therefore \quad r = 45^\circ$

At point M

$_1^2\mu = \frac{\sin r}{\sin i} = \frac{\sin 45^\circ}{\sin i} = \frac{1}{\sqrt{2}\cdot\sin i}$ **...(i)**

In ΔPMB

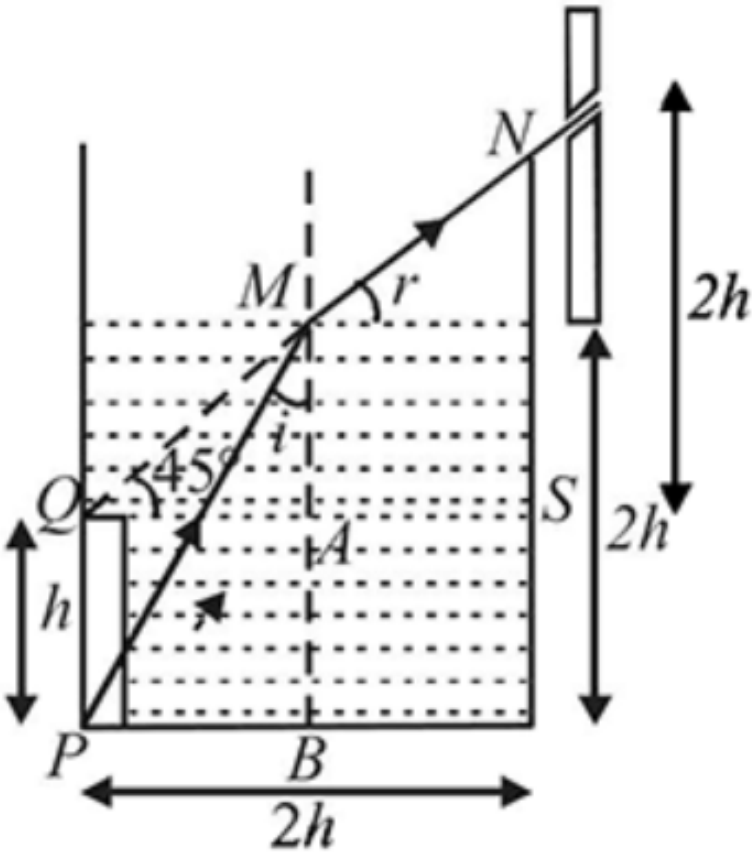
$PM^2 = 4h^2 + h^2 = 5h^2$

$\therefore \quad \sin i = \frac{h}{\sqrt{5}h}$

From (i) and (ii)

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$_2^1\mu = \sqrt{\frac{5}{2}}$



Q.9 Answer:

$\frac{(\mu-1)\mathrm{d}}{\mu}$

Solution:

Since $\frac{\text{Apparent depth}}{\text{Real depth}} = \frac{1}{\mu} \Rightarrow$ **Apparent depth** $= \mathrm{d}/\mu$

So mark raised up = Real depth - Apparent depth $= \mathrm{d} - \frac{\mathrm{d}}{\mu} = \mathrm{d}\left(1 - \frac{1}{\mu}\right) = \left(\frac{\mu-1}{\mu}\right)\mathrm{d}.$

Q.10 Answer:

10 cm

Solution:

Considering refraction at the curved surface,

$$\mathrm{u} = -20, \mu_2 = 1$$
$$\mu_1 = 3/2, \mathrm{R} = +20$$

Applying $\frac{\mu_2}{\mathrm{v}} - \frac{\mu_1}{\mathrm{u}} = \frac{\mu_2 - \mu_1}{\mathrm{R}}$

$\Rightarrow \frac{1}{v} - \frac{3/2}{-20} = \frac{1-3/2}{20} \Rightarrow v = -10$

i.e., 10 cm **below the curved surface or** 10 cm **above the actual position of flower.**

Q.11 Answer:

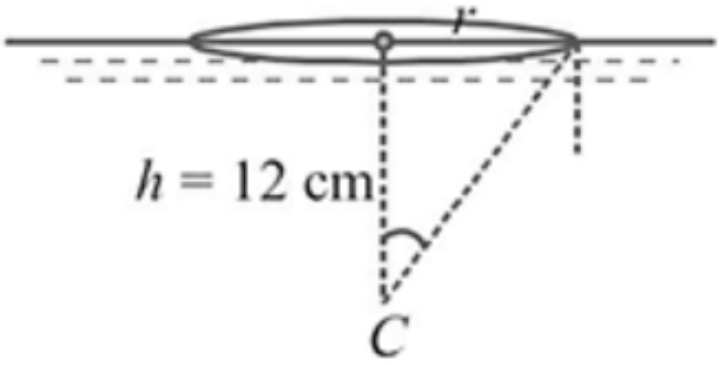
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13.6

Solution:

Now $r = h \tan C$

$= 12 \times \frac{3}{\sqrt{7}} = \frac{36}{\sqrt{7}} = 13.606 \text{ cm}$

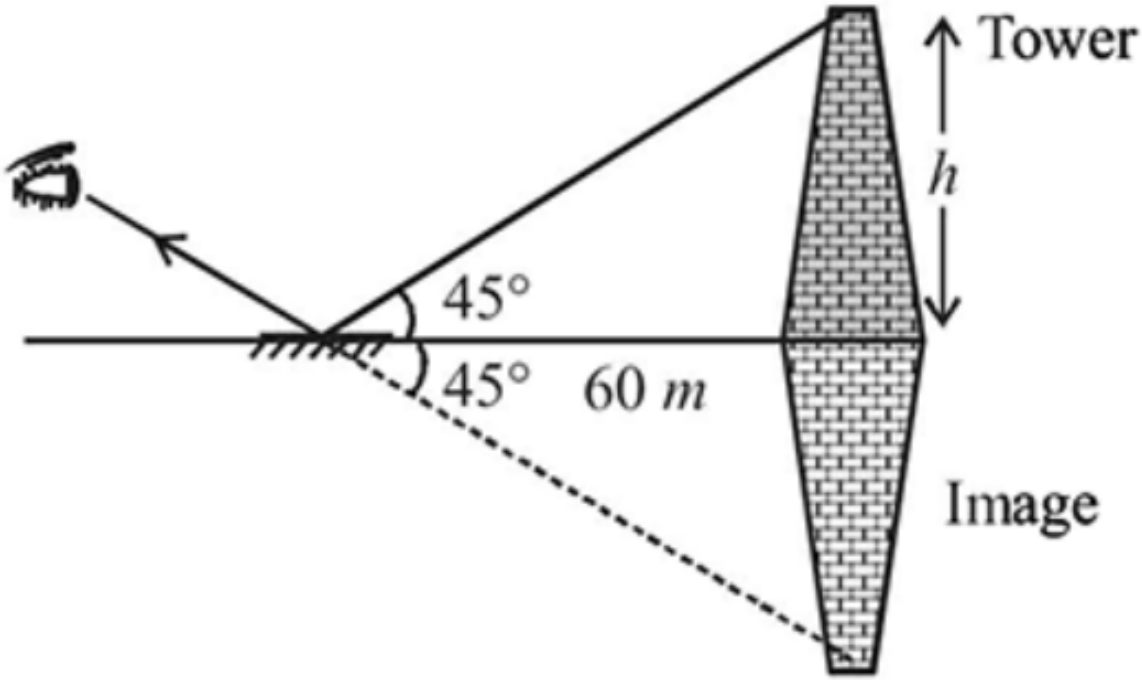


Q.12 Answer:

60

Solution:

$\tan 45^\circ = \frac{h}{60} \Rightarrow h = 60 \text{ m}$



Q.13 Answer:

$\mu_4 = \mu_1$

Solution:

Ray inside medium AB **is parallel to ray inside medium** CD

$\therefore$

Q.14 Answer:

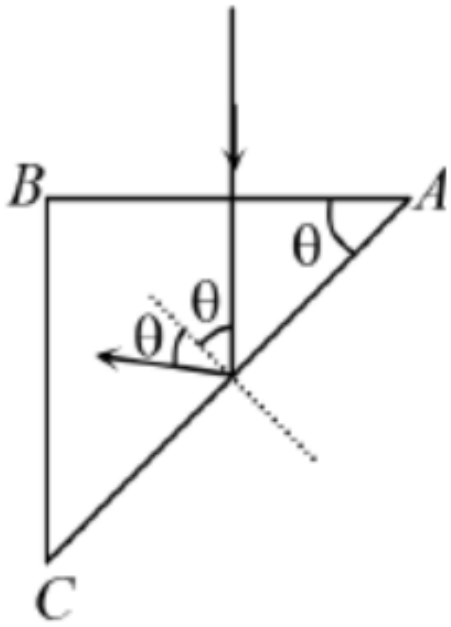
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$\sin\theta \geq \frac{8}{9}$

Solution:

$\sin\theta > \sin\theta_C = \frac{1}{\mu}$

$\sin\theta > \frac{1}{\frac{3/2}{4/3}}, \sin\theta > \frac{8}{9}$



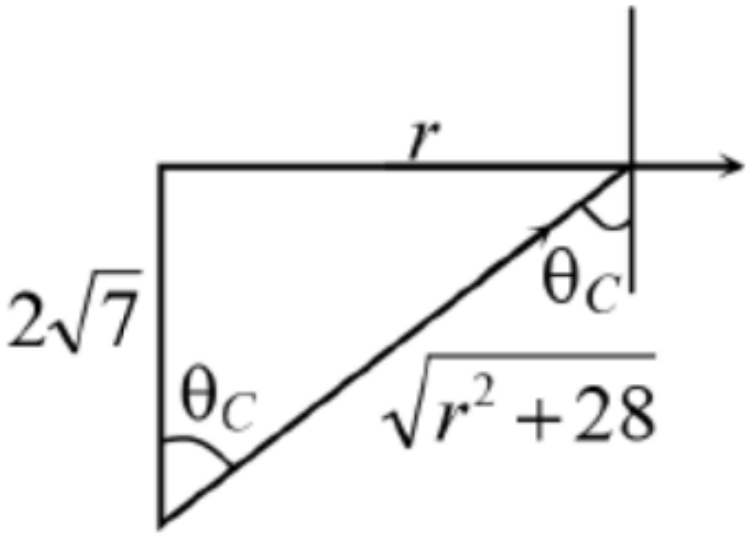
Q.15 Answer:

6 m

Solution:

$\sin\theta_C = \frac{1}{\mu} = \frac{r}{\sqrt{r^2+28}}$

$\mu = \frac{4}{3} \Rightarrow r = 6 \text{ m}$



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